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May 15, 2024
NRC-24-0025

10 CFR 50.73

U.S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, DC 20555-0001

Fermi 2 Power Plant
NRC Docket No. 50-341
NRC License No. NPF-43

Subject: Licensee Event Report (LER) No. 2024-001

Pursuant to 10CFR50.73(a)(2)(iv)(A), DTE Electric Company (DTE) is submitting LER No. 2024-001, Automatic RPS SCRAM on High RPV Pressure While Attempting to Lower Generator Output During RF-22.

No new commitments are being made in this submittal.

Should you have any questions or require additional information, please contact Mr. Eric Frank, Manager – Nuclear Licensing, at (734) 586-4772.

Sincerely,

A handwritten signature in black ink, appearing to be "P. Dietrich", written over a large, stylized circular flourish.

Peter Dietrich
Senior Vice President and Chief Nuclear Officer

Enclosure: Licensee Event Report No. 2024-001, Automatic RPS SCRAM on High RPV Pressure While Attempting to Lower Generator Output During RF-22

cc: NRC Project Manager
NRC Resident Office
Regional Administrator, Region III

**Enclosure to
NRC-24-0025**

**Fermi 2 NRC Docket No. 50-341
Operating License No. NPF-43**

**Licensee Event Report (LER) No. 2024-001
Automatic RPS SCRAM on High RPV Pressure While Attempting to
Lower Generator Output During RF-22**

NRC FORM 366 (04-02-2024)	U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104	EXPIRES: 04/30/2027
 LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)		Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov , and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.	

1. Facility Name Fermi 2	☒ 050 ☐ 052	2. Docket Number 00341	3. Page 1 OF 3
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4. Title Automatic RPS SCRAM on High RPV Pressure While Attempting to Lower Generator Output During RF-22

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
03	23	2024	2024	- 001 -	00	05	15	2024	NA		
									Facility Name	☐ 052	Docket Number
									NA		

9. Operating Mode 1	10. Power Level 023
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11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)					
☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		
☐ OTHER (Specify here, in abstract, or NRC 366A).					

12. Licensee Contact for this LER	
Licensee Contact Eric Frank - Manager, Nuclear Licensing	Phone Number (Include area code) 734-586-4772

13. Complete One Line for each Component Failure Described in this Report									
Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	Jl	68	X999	YES					

14. Supplemental Report Expected					15. Expected Submission Date			Month	Day	Year
☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)									

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)
At 0004 EDT on March 23, 2024, with the unit in Mode 1 at 23-percent power, the reactor automatically scrammed due to high reactor Pressure Vessel pressure when the turbine bypass valves unexpectedly closed while attempting to lower generator output to 55 MWe (per Fermi 2 procedure 22.000.04 "Plant Shutdown from 25-percent Power") to support shutdown for a refueling outage. The SCRAM was not complex, with systems responding normally post-scrum, with the exception of the pressure control system. The transient occurred while lowering Turbine Speed/Load Demand which caused a rise in pressure/power until the Reactor Pressure System Setpoint for Reactor Pressure High was exceeded and resulted in an Automatic Reactor SCRAM. Reactor water level was maintained at normal level following expected post-scrum feedwater level response. Decay heat was removed by the Main Steam System to the Main Condenser using manual operation of the turbine bypass valves. All Control Rods were inserted into the core. The cause of the SCRAM was the failure of the RL16 Block Load Relay within the turbine control system which resulted in a trip of a bypass valve logic and prevented the opening of the bypass valves, in automatic, when required. Fermi 2 has replaced the RL16 Block Load Relay and will modify several plant procedures.



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME		☒ 050 ☐ 052	2. DOCKET NUMBER	3. LER NUMBER		
Fermi 2			00341	YEAR	SEQUENTIAL NUMBER	REV NO.
				24	001	00

NARRATIVE

INITIAL PLANT CONDITIONS

Mode – 1
Reactor Power – 023

There were no structures, systems, or components (SSCs) that were inoperable at the start of this event that contributed to this event.

DESCRIPTION OF THE EVENT

At 0004 Eastern Daylight Time (EDT) on March 23, 2024, with the unit in Mode 1 at 23-percent power, the reactor automatically scrammed due to high Reactor Pressure Vessel (RPV) pressure when the turbine bypass valves [PCV] (which were being opened per Fermi procedure 22.000.04 "Plant Shutdown from 25-percent Power") unexpectedly closed while attempting to lower Generator [GEN] output to 55 MWe to support shutdown for a refueling outage. The SCRAM was not complex, with systems responding normally post-scrum, with the exception of the pressure control system. The transient occurred while lowering Turbine Speed/Load Demand which caused arise in pressure/power until the Reactor Pressure System (RPS) [JD] Set point for Reactor Pressure High was exceeded and resulted in an Automatic Reactor Scram.

Reactor water level was maintained at normal level following expected post-scrum feedwater level response. Decay heat was removed by the Main Steam System [SB] to the main condenser using manual operation of turbine bypass valves. All Control Rods were inserted into the core.

DESCRIPTION OF THE SYSTEM

The Main Turbine Bypass System [JI] and Moisture Separator Reheater (MSR) [RHTR] are designed to control steam pressure when reactor steam generation exceeds turbine requirements during unit startup, sudden load reduction, and cool down. It allows excess steam flow from the reactor to the condenser without going through the turbine. The bypass capacity of the Main Turbine Bypass System and Moisture Separator Reheater is 30-percent of the Nuclear Steam Supply System [SB] rated steam flow. Sudden load reductions within the capacity of the steam bypass can be accommodated without reactor SCRAM. The Main Turbine Bypass System consists of two valves connected to a 52-inch manifold, which is between the main steam isolation valves and the turbine stop valves. Each of these valves is operated by a separate hydraulic unitized actuator. The bypass valves are controlled by the Main Turbine Pressure Regulator Control System [JI], as discussed in the Updated Final Safety Analysis Report (UFSAR), Section 7.7.1.4. The bypass valves are normally closed, and the pressure regulator controls the turbine control valves that direct all steam flow to the turbine. If the turbine speed controls or the load limiter restricts steam flow to the turbine, the pressure regulator controls the system pressure by opening the bypass valves. When the bypass valves open, the steam flows from the 52-inch manifold through connecting piping to the condenser.

The reheating steam flow path to the moisture separator reheater provides an additional steam volume that mitigates a rapid pressure increase (e.g., from a generator load reduction).

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME Fermi 2	☒ 050	2. DOCKET NUMBER 00341	3. LER NUMBER		
	☐ 052		YEAR 24	SEQUENTIAL NUMBER 001	REV NO. 00

NARRATIVE**SIGNIFICANT SAFETY CONSEQUENCES AND IMPLICATIONS**

Due to the reactor protection system actuation while critical, this event was reported as a four-hour, non-emergency notification (EN 57046) per 10 CFR 50.72(b)(2)(iv)(B). Additionally, we received expected isolations for Level 3: Group 13 Drywell Sumps, Group 15 Traversing In-core Probes (TIPs) (which was already isolated) and Group 4 Residual Heat Removal – Shutdown Cooling (RHR-SDC) (which was already isolated). The Primary Containment Isolation Event is being reported under 10 CFR 50.72(b)(3)(iv)(A). Also, due to the main turbine bypass valves unexpectedly closing, this was initially reported under 10 CFR 50.72(b)(3)(v)(D), but later revised to remove this reporting criteria because this was not an event that would have prevented the fulfillment of a safety function.

Unexpected bypass valve closure during operation of turbine controls speed/load demand resulted in reactor pressure increase. This created a positive feedback loop where the pressure rise raises reactor power which further raises pressure. The subsequent turbine control valve closure over a one-minute period accelerated the reactor pressure increase to above the reactor pressure SCRAM set point of 1093 psig and an automatic Reactor SCRAM occurred. The maximum reactor pressure reached was 1102 psig. The plant response was assessed against the descriptions provided in UFSAR Section 15.2 "Increase in Reactor Pressure", as symptoms of event are similar. While UFSAR analyses typically assume 100-percent power operation, the response of the plant to increasing pressure at the power level of this event [22.5-percent Core Thermal Power (CTP)] is consistent with Fermi's design features. Reactor Pressure - High RPS trip functioned at the correct set point and plant was recovered using feedwater and the Main Condenser as a heat sink.

As such, there was no impact to the health and safety of the public.

CAUSE OF THE EVENT

The cause of the SCRAM was the failure of the RL16 Block Load Relay which resulted in a trip of a bypass valve logic and prevented the opening of the bypass valves, in automatic, when required. The RL16 relay provides a signal to add 55 MWe (load) to the Narrow Range Speed Governor (NRSG) signal when the unit is connected to the grid. NRSG is one of 5 modes of operation of the turbine control system that maintains the speed of the turbine at 1800 RPM or the load of the main generator at a given setpoint. The remaining 4 modes are Pressure, CVOL (Turbine Flow Limit), Run Up, and Turbine Trip. Of these 5 control signals, the lowest signal, which corresponds to the lowest valve demand, is selected to be in control by a low value gate. The output of the low value gate is sent to control the steam inlet valves to the turbine. The bypass valves are positioned by the difference in the output of the low value gate and the pressure signal to maintain pressure at the 52-inch manifold. In normal operation, the pressure control signal is maintained as the low value to the low value gate and is in control.

CORRECTIVE ACTIONS

Fermi 2 has replaced the RL16 Block Load Relay.

Fermi Procedure 22.000.04 ("Plant Shutdown from 25-Percent Power") Section 6 will be revised to eliminate operating on Narrow Range Speed Governor (NRSG) during reactor shutdown due to the potential latent failure mechanism.

PREVIOUS OCCURRENCES

There are no previous occurrences.